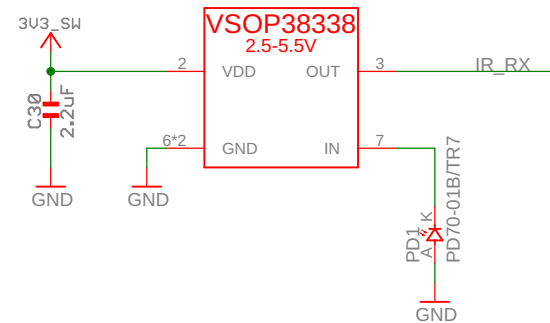
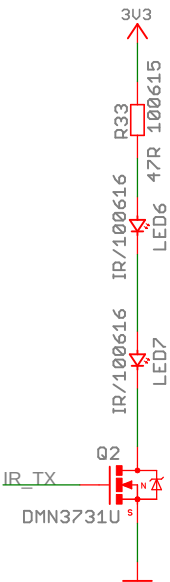


Github Badge 2026

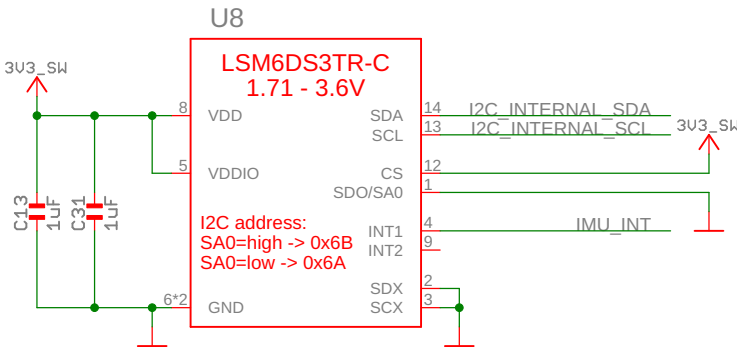
IR Receiver



IR Transmitter

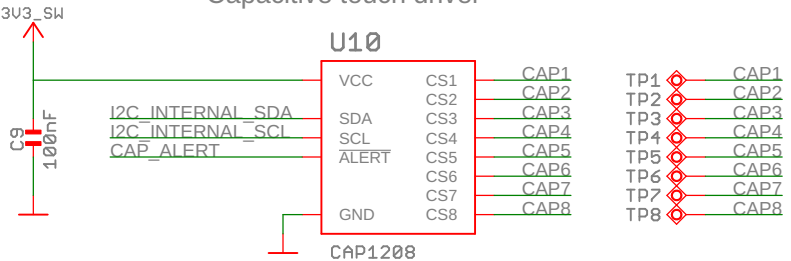


LSM6DS3TR-C [3D accelerometer and 3D gyroscope]



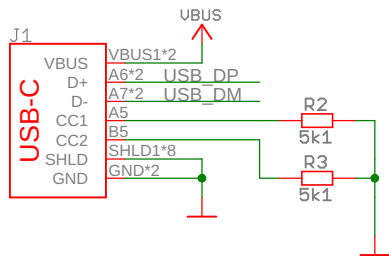
LSM6DS3TR-C (note: this is different from LSM6DS3TR)

Capacitive touch driver

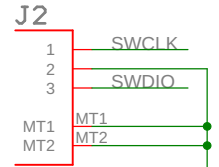


Github Badge 2026: RP2350 & co

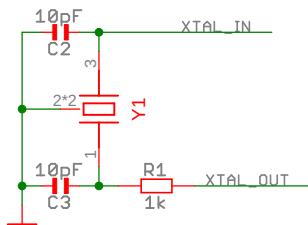
USB-C connector



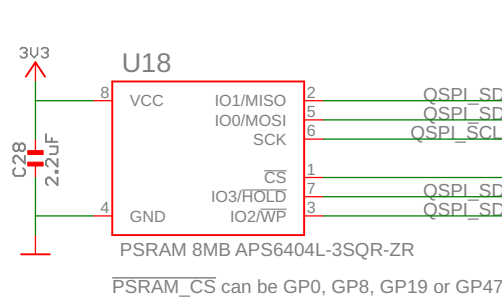
Debug Connector



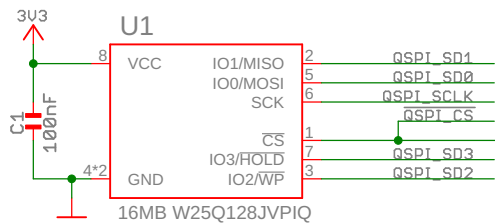
Crystal



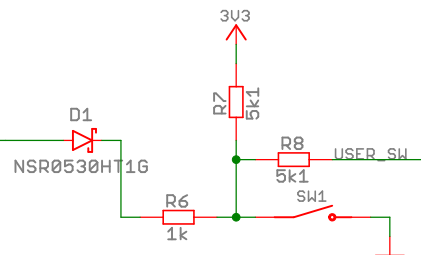
PSRAM



Flash memory



Combined USB boot & user switch

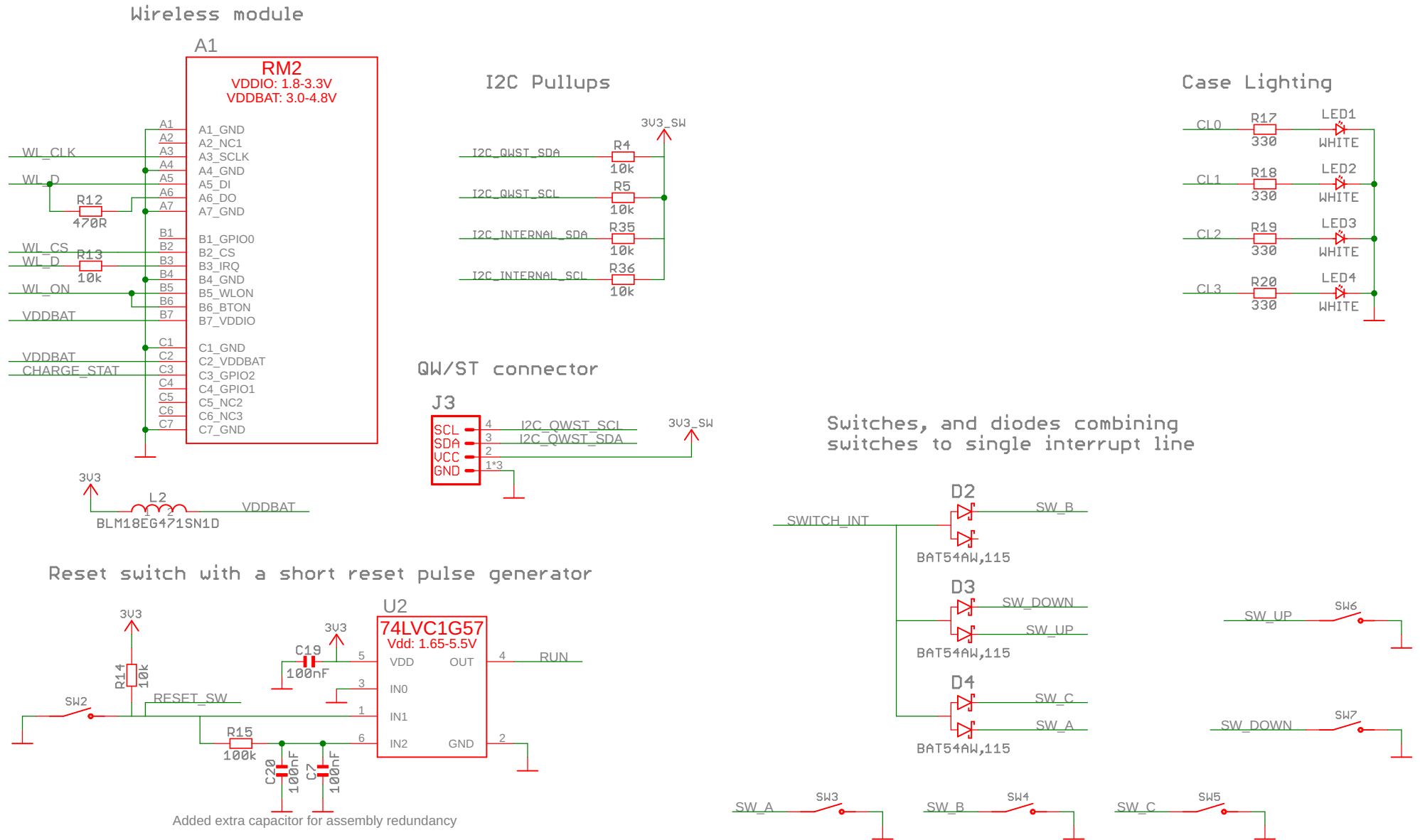


MCU1

RP2350B
IOVDD&VREG: 1.8-3.3V
USB&ADC: 3.3V
DVDD: 1.1V

ADC_IOVDD	GPIO0	77	CL0
	GPIO1	78	CL1
	GPIO2	79	CL2
	GPIO3	80	CL3
	GPIO4	1	I2C_QWST_SDA
	GPIO5	2	I2C_QWST_SCL
USB_OTP_IOVDD	GPIO6	3	SW_DOWN
QSPI_IOVDD	GPIO7	4	SW_A
VREG_VIN			
	GPIO8	6	PSRAM_CS
	GPIO9	7	SW_B
	GPIO10	8	SW_C
	GPIO11	9	SW_UP
VREG_AVDD	GPIO12	11	VBUS_DETECT
	GPIO13	12	CAP_ALERT
	GPIO14	13	RESET_SW
	GPIO15	14	SWITCH_INT
VREG_LX			
	GPIO16	16	IR_TX
VREG_FB	GPIO17	17	IR_RX
DVDD	GPIO18	18	I2C_INTERNAL_SDA
	GPIO19	19	I2C_INTERNAL_SCL
	GPIO20	20	IMU_INT
	GPIO21	21	LCD_VSYNC
	GPIO22	22	USER_SW
	GPIO23	23	WI_ON
XIN	GPIO24	25	WI_D
XOUT	GPIO25	26	WI_CS
	GPIO26	27	LCD_BACKLIGHT
	GPIO27	28	LCD_CS
RUN	GPIO28	36	LCD_RS
	GPIO29	37	WI_CLK
SWDIO	GPIO30	38	LCD_WR
SWCLK	GPIO31	39	LCD_RD
QSPI_SD0	GPIO32	40	LCD_DB0
QSPI_SD1	GPIO33	42	LCD_DB1
QSPI_SD2	GPIO34	43	LCD_DB2
QSPI_SD3	GPIO35	44	LCD_DB3
QSPI_SCLK	GPIO36	45	LCD_DB4
QSPI_CS	GPIO37	46	LCD_DB5
	GPIO38	47	LCD_DB6
	GPIO39	48	LCD_DB7
USB_DP	GPIO40/ADC0	49	VBAT_SENSE
USB_DM	GPIO41/ADC1	52	SW_POWER_EN
	GPIO42/ADC2	53	
	GPIO43/ADC3	54	LIGHT_SENSE
	GPIO44/ADC4	55	
	GPIO45/ADC5	56	
	GPIO46/ADC6	57	
	GPIO47/ADC7	58	
VREG_PGND			
GND			

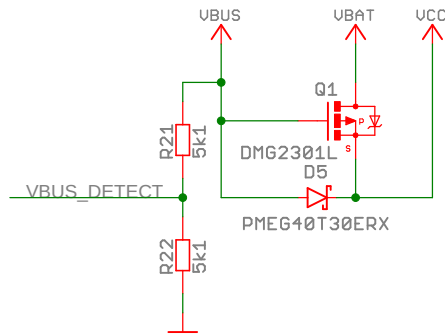
Github Badge 2026: Wireless, switches, LEDs, ...



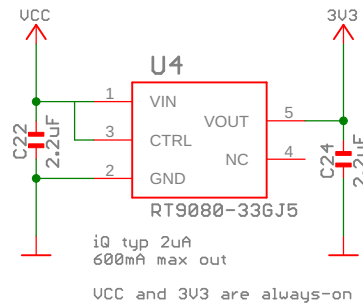
Github Badge 2026: Power

Power Supply and switching

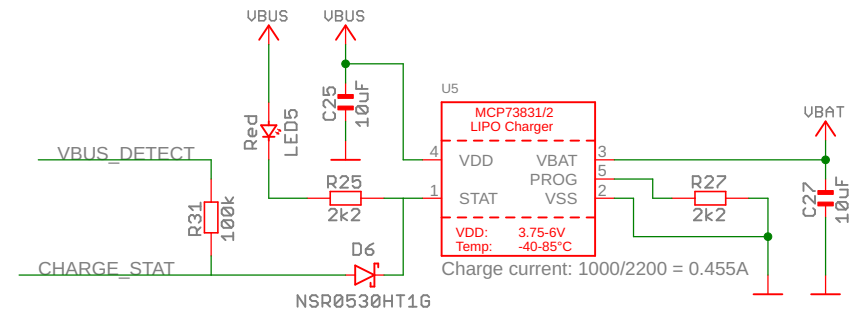
Use VBUS when available otherwise VBAT for supply.



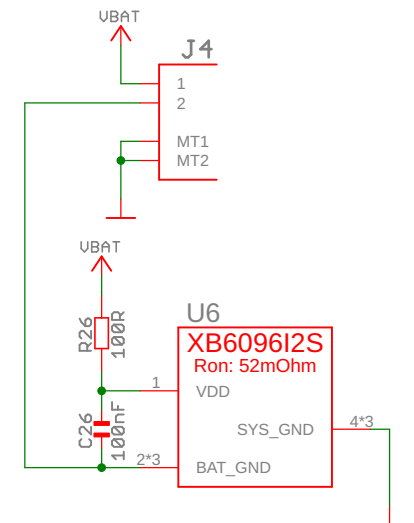
LDO regulator



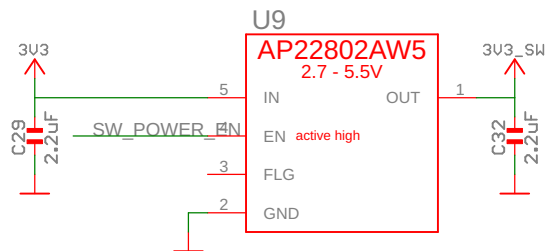
Battery charger



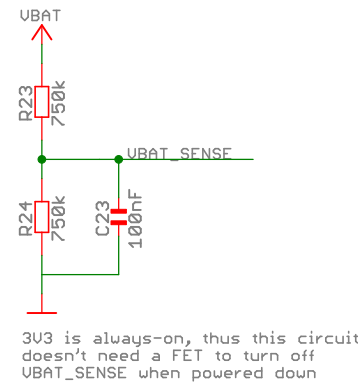
Battery connector (Picoblade) and protection



Switchable 3v3

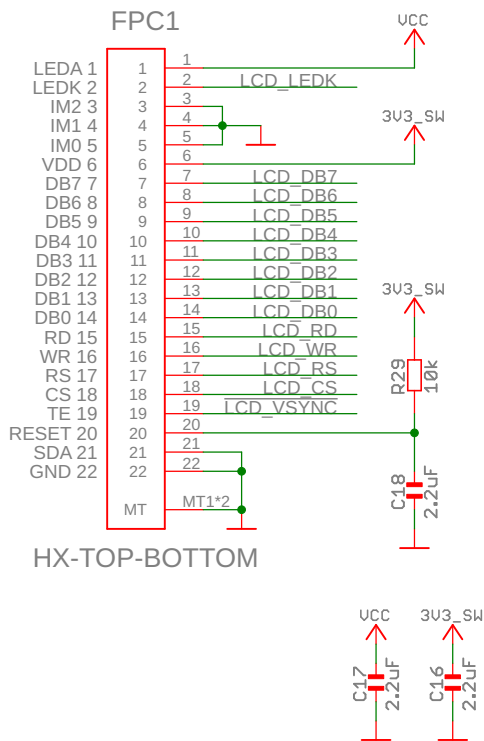


Battery voltage sense



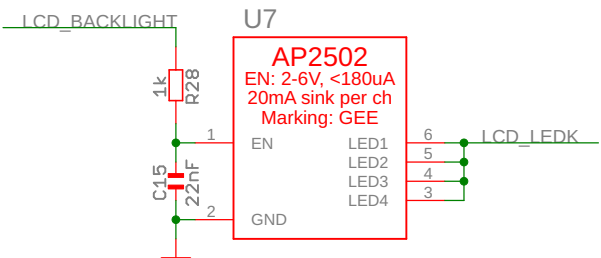
Github Badge 2026: LCD Display and Light Sensor

LCD connector & circuitry



LCD Backlight

Display backlight is 4 white LEDs in parallel (FU 3.2 - 20mA). Each channel of the AP2502 constant current driver sinks 20mA so we tie them together to provide the necessary current.



Light sensor

